

# CAGE Drones Digital Platform

An integrated training, operations, and business management platform for professional drone certification, flight logging, and client engagement.

120+

Surveys completed

50k+

Hectares mapped

80+

Pilots trained

20+

Enterprise clients

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## Executive Summary

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CAGE Drones has built a purpose-designed digital platform that unifies public marketing, pilot training, flight operations, and back-office administration in a single secure web application.

The platform supports the full lifecycle of a professional drone training organisation — from the moment a prospective client discovers CAGE online, through structured classroom and field training, to certificate issuance and public verification. It is tailored to CASA-standard Remote Pilot License (RePL) programmes and specialised industrial pathways including mapping, inspection, and precision agriculture.

**Key value proposition:** One platform replaces disconnected spreadsheets, email threads, and paper logbooks with a modern, role-based system that scales training delivery while giving administrators real-time visibility across students, flights, bookings, and business inquiries.

## Who the Platform Serves

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### Prospective Clients & Trainees

- Browse certification programmes and pricing
- View live training schedules and cohort availability
- Submit service bookings and quotation requests
- Apply for upcoming training cohorts
- Verify certificates online without an account

### Students (Pilots in Training)

- Access enrolled courses and learning resources
- Complete timed theory exams and assignments
- Track grades, feedback, and progress
- Maintain a digital flight logbook with checklists
- Download certificates of completion

### Teachers (Instructors)

- Create and manage courses and cohorts
- Build exams, grade assignments, and run gradebooks
- Review student flights and provide structured feedback
- Maintain the aircraft register
- Post course announcements

### Administrators

- Manage users and role assignments
- Process bookings, quotations, and applications
- View platform-wide reports and analytics
- Monitor email notifications and outbox
- Oversee all training and operational data

# Platform Architecture

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The application is a modern, cloud-hosted web platform built for reliability, security, and mobile access. It uses industry-standard technologies and stores all data in a secure PostgreSQL database with row-level access controls — ensuring each user sees only what they are authorised to access.

| Component           | Description                                                                                   |
|---------------------|-----------------------------------------------------------------------------------------------|
| Web application     | Responsive Next.js application accessible from desktop and mobile browsers                    |
| Authentication      | Secure email-and-password sign-in with role-based portal routing                              |
| Database & storage  | Supabase (PostgreSQL) with encrypted file storage for course materials and assignment uploads |
| Email notifications | Automated admin alerts for bookings, applications, and quotations (via Resend)                |
| Hosting             | Production deployment on Render with optional Vercel support                                  |

# Public Website & Lead Generation

The public-facing homepage serves as CAGE's digital storefront. It presents the company's training credentials, client portfolio, technology fleet, and team — while converting interest into actionable leads.

## Certification Programmes

**Remote Pilot License (RePL)**

Entry Level

5 Days · \$1,950

Foundational certification for commercial drone operations. Covers air law, flight theory, and practical assessment to CASA standards.

**Specialized Mapping**

Specialist

3 Days · \$1,400

Advanced photogrammetry and LiDAR data acquisition for high-precision surveying and digital twin creation.

**Advanced Inspection**

Advanced

· \$2,850

Critical infrastructure inspection including thermal imaging, structural analysis, and close-proximity flight in high-interference environments.

## Inbound Business Flows



## Notable Public Features

- **Live training schedule** — cohort dates, locations, and availability (open / waitlist) pulled from the database
- **Service booking form** — capture site location, preferred dates, service type, and project area
- **Quotation requests** — structured project scope, deliverables, timeline, and budget
- **Certificate verification** — anyone can verify authenticity at */verify* using the certificate serial number
- **Client portfolio** — documented engagements with UNICEF, ESCOM, Illovo Sugar, SMEC, and others

## Student Learning Portal

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Enrolled students access a dedicated learning hub where all training activity is centralised. The portal is designed for clarity and mobile use, with a sidebar navigation covering every aspect of the training journey.

| Module           | Capabilities                                                                                                 |
|------------------|--------------------------------------------------------------------------------------------------------------|
| Dashboard        | Course overview, announcements, upcoming deadlines, exam availability, and performance summary               |
| Exams            | Timed multiple-choice assessments with automatic scoring against a configurable pass threshold (default 70%) |
| Assignments      | View instructions, upload deliverables, and receive graded feedback from instructors                         |
| Grades & Results | Consolidated view of exam scores and assignment outcomes                                                     |
| Resources        | Access PDFs, videos, slides, and links with progress tracking                                                |
| Flight Logbook   | Digital pilot log with pre/post-flight checklists, weather, GPS location, and mission details                |
| Certificates     | View and download issued certificates of completion                                                          |

### Digital Flight Logbook

A distinguishing feature of the CAGE platform is its integrated flight operations module. Students log every training flight with structured data that mirrors professional aviation practice:

- Aircraft assignment from the organisation's register
- Mission type, location, and GPS coordinates
- Departure/arrival times, duration, and weather conditions
- **Pre-flight checklist** — airspace authorisation, NOTAMs, site survey, battery checks, compass calibration, RTH altitude, emergency briefing
- **Post-flight checklist** — power-down, damage inspection, log security, anomaly documentation, storage
- Status workflow: Planned → In Progress → Completed (or Cancelled)
- Instructor review status: Unreviewed → Approved / Needs Attention

## Teacher & Instructor Portal

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Instructors have a full course management toolkit plus operational tools for flight supervision and fleet administration.

## Per-Course Tools

### Course Management

Overview, student enrollment, announcements, and publishing controls.

### Assessment

Create assignments and exams; review submissions; maintain a gradebook.

### Learning Resources

Upload and organise course materials in secure cloud storage.

### Exam Builder

Author question banks, set time limits and pass marks, publish when ready.

## Operations Tools

- **Flight reviews** — instructors score flights on preflight, airmanship, procedures, decision-making, and postflight (1–5 scale); assign overall grades and written feedback
- **Aircraft register** — fleet inventory with registration, model, manufacturer, serial number, and max takeoff weight
- **Training cohorts** — schedule management with codes, dates, locations, capacity, and status (open / waitlist / closed)

## Administration Console

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Platform administrators have a dedicated operations dashboard with full visibility across users, learning activity, flight operations, and business pipeline.

| Section             | Purpose                                                                                          |
|---------------------|--------------------------------------------------------------------------------------------------|
| Dashboard           | Real-time counts: users, courses, exams, enrollments, resources; role breakdown; recent sign-ups |
| Users               | Assign roles (Student, Teacher, Admin); manage access across all portals                         |
| Inquiries           | General enrollment interest submissions                                                          |
| Bookings            | Service requests — status tracking from pending through confirmed to completed                   |
| Quotations          | Project quote requests with amount, currency, and pipeline status                                |
| Applications        | Training cohort applications — review, accept, waitlist, or decline                              |
| Reports & Analytics | Cross-platform metrics: users, learning outcomes, flights, certificates, business pipeline       |
| Email Outbox        | Audit trail of notification emails (pending, sent, failed)                                       |

## Certificates & Trust

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Upon successful completion of a course, instructors or administrators can issue a **Certificate of Completion** with a unique serial number, grade, and issue date. Certificates can be revoked if needed, with reason and timestamp recorded.

A public verification page allows employers, regulators, or clients to confirm certificate authenticity by entering the serial number — without requiring a login. Only non-sensitive information is exposed: student name, course title, issue date, grade, and revocation status.

## Security & Data Protection

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- **Role-based access** — three distinct roles with enforced portal separation
- **Row-level security** — database policies ensure users access only their own data (students) or data for courses they manage (teachers)
- **Immutable exam attempts** — once submitted, exam answers cannot be altered
- **Role change protection** — only administrators can change user roles; enforced at the database level

- **Private file storage** — course resources and assignment uploads stored in secure, access-controlled buckets
- **No exposed secrets** — the application uses only public API keys in the browser; administrative database access is server-side only

## Technology Fleet & Expertise

CAGE's platform reflects the organisation's operational capabilities across enterprise drone hardware and software:

DJI Matrice 300/350 RTK · DJI Mavic 3 Multispectral · DJI Mavic 3 Enterprise · DJI Mavic 3 Thermal · DJI Agras T50 · Zenmuse L1 LiDAR · Pix4Dmapper · Pix4Dfields · DJI Terra · QGIS · AutoCAD

### Selected Client Engagements

| Client                            | Scope                                                                                  |
|-----------------------------------|----------------------------------------------------------------------------------------|
| UNICEF Malawi                     | ~19,700 ha flood resilience mapping — 5 cm orthomosaics, DSM/DTM, GIS flood modelling  |
| ESCOM Malawi                      | 48 engineers — 3-week RPL programme and advanced mapping with LiDAR corridor modelling |
| Illovo Sugar Malawi               | 17 staff trained for RePL and DJI Matrice 350 RTK familiarisation                      |
| SMEC                              | Lower Domasi Dam feasibility — ~1,207 ha surveyed with M300 RTK + LiDAR                |
| Paramount Holdings / Salima Sugar | Precision aerial spraying with DJI Agras T50 across sugarcane estates                  |



# Platform Modules at a Glance

| Domain            | Features                                                                                                                      |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Marketing & CRM   | Public website, live schedule, service bookings, quotations, training applications, enrollment inquiries, email notifications |
| Learning (LMS)    | Courses, enrollments, resources, announcements, assignments, exams, gradebook, progress tracking                              |
| Flight Operations | Aircraft register, pilot logbook, pre/post checklists, instructor evaluations, aggregated flight hours                        |
| Credentials       | Certificate issuance, revocation, public serial verification                                                                  |
| Administration    | User management, business pipeline, analytics, email outbox audit                                                             |

## Deployment & Operations

The platform is production-ready and configured for cloud hosting with the following operational characteristics:

- Environment-based configuration for database, authentication, and email
- Automated build and deploy pipeline (Render Blueprint included in repository)
- Database migrations managed as versioned SQL scripts
- First administrator bootstrapped via secure one-time database promotion
- Optional inline email delivery (Resend) or queued outbox for background processing

**Summary:** The CAGE Drones platform is a complete digital backbone for a professional drone training and services organisation — combining marketing, education, flight compliance, credentialing, and business operations in one secure, scalable system built for the Malawi market and CASA-aligned training standards.